

My Journal

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Contributions:

- Ai.py
- Service.php (Derived from V1)
- Connection.php (Derived from V1)
- Clients (Derived from V1)

I made V1 programs compatible with the new architecture of V2. I mostly modified existing code and created new logic in ai.py. Take note that ai.py is a mock version intended to simulate what the real system could be. It uses a flag named 'raw' to determine whether the output is random or not. This is a simulation as it is supposed to send computed trends in raw format. I implemented the AI part with raw TCP sockets instead of using HTTP or WebSocket for simplicity. I chose to use PHP as a proxy to enhance security and make the server responsible for the AI logic instead of the clients, which would otherwise create a major security issue. I chose SQLite over MySQL that was used in V1 for simplicity. Also, like in V1, the server automatically initializes the database.

The biggest challenge I encountered was making existing code compatible with the new program. Changes were hard to implement, but I managed to do it. The next big challenge I faced was the buffer size problem. When sending data across TCP sockets, it needs to read all data sent, otherwise the data is corrupted and unusable, which can lead to data corruption and security issues.

I fixed the compatibility problem by modifying the source. The solution was straightforward. Then I fixed the buffer size problem using “\n” as a delimiter. The program would receive incoming data until it encounters “\n”, and then it would store the leftovers into “buffer”. It is not necessary, but I implemented it just in case.

<Relevant diagrams and images are in the next pages>

Using curl to communicate with the PHP API

```
Ubuntu
jhan@ZephyrusPTIII:~$ echo test service.php using curl
test service.php using curl
jhan@ZephyrusPTIII:~$ curl -X POST "http://localhost:8000/?action=generate_api"
{"status":"success","message":"API key generated successfully","data":{"api_key":"c880a1031918a75425d050ecb20022d5"}}
jhan@ZephyrusPTIII:~$ curl -X POST "http://localhost:8000/?action=execute_ai" -H "Content-Type: application/json" -d '{"api_key": "c880a1031918a75425d050ecb20022d5", "month": 20, "raw": true}'
{"status":"success","message":"AI Execution is successful!","data":[7,92,40,42,63,39,86,32,63,3,32,61,22,26,87,89,19,89,37,31]}
jhan@ZephyrusPTIII:~$ curl -X POST "http://localhost:8000/?action=get_all_api"
{"status":"error","message":"Endpoint not found","data":null}
jhan@ZephyrusPTIII:~$ curl -X POST "http://localhost:8000/?action=api_keys"
{"status":"error","message":"Endpoint not found","data":null}
jhan@ZephyrusPTIII:~$ curl -X GET "http://localhost:8000/?action=api_keys"
{"status":"success","message":"API keys retrieved successfully","data":[{"id":1,"api_key":"f70c84706eb99d2fd1a982412948044b","created_at":null},{"id":2,"api_key":"51ba113b9a10ee0204cc39c72cfd053","created_at":null},{"id":3,"api_key":"945703ee6954045cda40009379fd393f","created_at":null},{"id":4,"api_key":"a455846fd93bd66ad435898ebce43098","created_at":null},{"id":5,"api_key":"ddd2b651ecd6a5116f0592c5ede44852","created_at":null},{"id":6,"api_key":"6ce0929e14fe3c3b79fd1b856a8900a3","created_at":null},{"id":7,"api_key":"b4a130d990e35355fe052e46bc7f5541","created_at":null},{"id":8,"api_key":"0e29afd08236e6d2f95c23feb77d44c8","created_at":null},{"id":9,"api_key":"eedda6e0f8b9ee71df3d1192d3ad36d","created_at":null},{"id":10,"api_key":"64e0bbfa056d64408c36602a9ad23fdf","created_at":null},{"id":11,"api_key":"505a1cd72b2023ed9393892e67749352","created_at":null},{"id":12,"api_key":"c880a1031918a75425d050ecb20022d5","created_at":null}]}
jhan@ZephyrusPTIII:~$
```

Using netcat to communicate with the AI API

```
Ubuntu
jhan@ZephyrusPTIII:~$ echo testing ai.py using netcat
testing ai.py using netcat
jhan@ZephyrusPTIII:~$ echo '{"cmd": "ping"}' | nc 127.0.0.1 1234
{"response_code": 0, "response_message": "pong!"}
^C
jhan@ZephyrusPTIII:~$ echo '{"cmd": "run", "month": 12, "raw": false}' | nc 127.0.0.1 1234
{"response_code": 0, "response_message": "Data transmission success!", "data": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]}
^C
jhan@ZephyrusPTIII:~$ echo '{"cmd": "run", "month": 1, "raw": true}' | nc 127.0.0.1 1234
{"response_code": 0, "response_message": "Raw data transmission success!", "data": [80]}
^C
jhan@ZephyrusPTIII:~$ echo '{"cmd": "shutdown"}' | nc 127.0.0.1 1234
{"response_code": 2, "response_message": "Server shutting down. Bye!!!"}
jhan@ZephyrusPTIII:~$
```